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DFIT-Driven Formation Properties Impact on the Performance of Unconventional Wells with Consistent Drilling and Completion Practices

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Abstract

The scope of this paper is to investigate the impact formation properties, driven from diagnostic fracture injection test (DFIT), have on the performance of unconventional wells with identical drilling and completion practices. The objective is to study the relationship of closure time and transmissibility with well performance, to improve the decision-making process and performance prediction of future wells within an area of application.

The performance of unconventional wells is mainly driven by either formation related parameters such as rock quality, reservoir pressure, and fluid properties, or stimulation related parameters such as fracture geometry and complexity. Four scattered wells within the same application area with identical drilling and completion practices targeting the same unconventional formation were used for this study. The stimulation design was fixed for all four wells, leaving formation properties as the only variable. DFIT was implemented on all four wells to acquire key formation related properties, mainly closure time and transmissibility. These two parameters were then correlated with well performance to quantify its impact, and identify potential prediction techniques for future wells.

Closure time refers to the time required for a fracture to close after injection, while transmissibility refers to the ability of fluids to pass through rock with a certain height and fluid viscosity. The use of both parameters yielded key insights when plotted against wells productivity index (PI), especially that all four wells had approximate closure and pore pressure gradients. When closure time was plotted against PI, an inverse exponential correlation was matched with an accuracy of more than 99%, indicating that a longer closure time consistently yielded lower well performance. On the other hand, a transmissibility versus PI plot showed a direct exponential correlation with an accuracy of 98%, emphasizing on the positive impact transmissibility has on well performance. The combined use of closure time and transmissibility yielded definitive and consistent results across all four wells which can be used to drive conclusions on well evaluation, and navigate the decision-making process at the subject application areas.

The novelty of this work lies in the utilization of DFIT-driven parameters, mainly closure time and transmissibility, in evaluating the performance of unconventional wells within an area of application.

Correlations with well PIs were matched with a high degree of accuracy, indicative of the methodology's validity, and potential broader application.

Introduction

The definition of unconventional formations has been widely discussed in the industry, with practitioners often relating it to specific formation-related properties such as extremely low permeability, while others relate it to the intense stimulation requirements necessary to unlock its full potential. In reality, both perspectives are valid and interconnected. The low permeability of unconventional formations is effectively addressed by applying various stimulation techniques, creating relatively high permeable flow paths allowing the commercial flow of hydrocarbons. One of the most widely used stimulation techniques in unconventional resources is hydraulic fracturing, a process that includes the pumping of a mixture of fluids and solid particles at high pressures, breaking down targeting formation rock and creating effective hydrocarbon flow paths. In addition to stimulation, a strategic enabler unlocking the commerciality of unconventional resources is gaining a deeper understanding of targeted rock formations, through continuous diagnostics and technical evaluation. Figure 1 illustrates a summary of the most dominant drivers of unconventional well performance. These key drivers can be broadly categorized into rock quality (influenced mainly by formation permeability, geo-mechanical properties, petrophysical properties, and the existence of natural fracturs), reservoir pressure and fluid properties, and stimulation geometry and complexity. In recent years, a shift towards diagnostic pragmatism is observed through the deployment of non-invasive, operationally friendly techniques. A prominent example of such techniques is the Diagnostic Fracture Injection Test (DFIT), a post-injection pressure leak-off monitoring test that infers geo-mechanical and reservoir-related properties. DFIT involves injecting a small volume of fluid to create a small hydraulic fracture, which is then followed by a period of pressure monitoring that continues until the injected fluid is fully dissipated into formation. Pressure monitoring data from DFIT is typically divided into three timeframes that relate to the closure of the induced mini hydraulic fracture: pre-closure, closure, and post-closure. During the pre-closure phase, the induced fracture remains open, and the data collected during this period is used to calculate the Fracture Gradient (FG), which informs engineers about the maximum pressure required to cause rock failure. The closure phase provides estimates of the fracture closure pressure – or minimum stress – and the time it takes for the created fracture to close, a parameter that is related to the rock's permeability and the formation's fluid viscosity. The post-closure phase yields insights into formation-related properties, mainly the flow regime and reservoir pressure. Collectively, the data acquired from DFIT guides engineers and geoscientist to make informed decisions in the design and execution of unconventional well placement, drilling, stimulation, and production. This paper builds on the lessons learned from DFIT and focused on developing a predictive workflow that leverages DFIT-driven results to qualitatively forecast the performance of unconventional wells post-stimulation. The methodology was applied on multiple wells with consistent drilling and completion practices to ensure a fair and unbiased assessment, with a total of four application examples investigated. The first application example was conducted in a controlled area with similar subsurface properties, which revealed the existence of a correlation between DFIT-driven properties and well performance. The second application example, performed in another area, demonstrated the repeatability of the identified correlation. On the other hand, the third and fourth application examples investigated limitations of the identified correlation, emerging from variations in reservoir pressure or targeted layers within the same area of study.

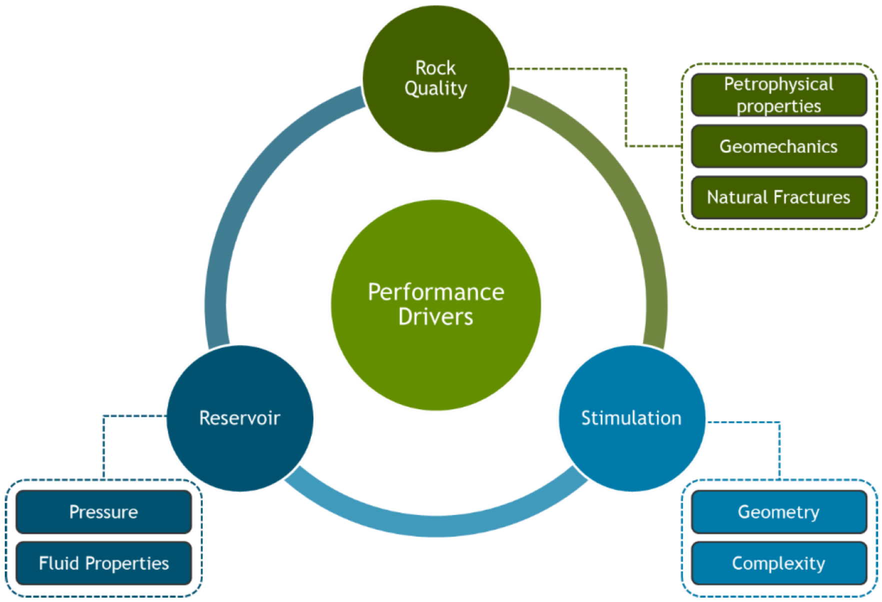

Figure 1—Drivers of unconventional well performance (rock quality, reservoir properties, and stimulation design)

Background

Unconventional resources that are found in source rocks, are formed under unique depositional environment that shape their characteristics. These resources accumulate under low energy settings over extensively long periods of time, ultimately yielding higher uniformity in subsurface properties within a localized area. This paper investigated wells that targeted an unconventional source rock, located in areas that have relatively high degree of spatial continuity in subsurface properties. These subject wells had consistent drilling and completion designs to ensure that their deliverability was not influenced by human-induced controllable parameters. All wells followed a standardized operational workflow that started with drilling, followed by pre-stimulation diagnostics using DFIT, then stimulation, and finally post-stimulation well testing. The final step (post-stimulation well testing) exhibited different results across wells within the same area. This ignited the interest in developing a predictive approach that could qualitatively forecast well performance early in the well life cycle, potentially informing the decision-making process, which might influence well development strategies. DFIT was selected as the primary means of diagnostics, and was performed consistently and uniformly on all wells. From the three pre-defined DFIT timeframes (pre-closure, closure, and post-closure), three parameters were selected for comparative analysis. The first is closure gradient, which served as an analog for the homogeneity in stresses between wells, ensuring that targeted formations across all wells had similar geo-mechanical properties. The second parameter is closure time, which represented the time required for the induced hydraulic fracture to close, providing insights into formation transmissibility (permeability and fluid viscosity). The third parameter is the reservoir pressure gradient, estimated from post-closure analysis, which provided insights into the reservoir's capability to deliver and produce. From these key parameters, closure time was selected as the analog for comparison between wells, which will be used to identify potential correlations with post-stimulation well performance. The rationale behind using closure time as a predictive parameter is justified by its relationship with formation transmissibility, a parameter that accounts for the reservoir's permeability, height, and fluid viscosity. Although fluid efficiency, which is the ratio of pumped fluid volume stored inside the fracture over the total volume of fluid pumped (S&P Global Commodity Insights, 2024), can also be considered as a reference point for correlating well performance due to its similar relationship with transmissibility, closure time yielded a wider range of variability, which ultimately resulted in a higher degree of accuracy in predicting well performance. From post-stimulation well testing, productivity index (PI) was selected as the key correlating parameter. PI is a metric that quantifies the volume of hydrocarbon production per pressure unit. It was selected to normalize and account for the variability in production parameters between wells at the early phase of production, such as production rates, bottom-hole well flowing pressures, and the reservoir pressure. These two parameters, closure time and PI, established the foundations of this study and the outcomes that followed.

Methodology

This main focus of this paper was to investigate the relationship between DFIT-acquired formation properties and well performance. The approach relied on two key parameters: fracture closure time and PI. To maintain objectivity, the workflow utilized data from a set of unconventional wells targeting the same formation, with identical drilling and completion practices. DFIT was implemented in a standard manner, with injection volume and rate, as well as monitoring durations maintained constant, to acquire comparable fracture and formation-related properties for all wells. Fracture closure time was selected as the primary parameter to correlate with well productivity, as it serves as an effective proxy for reservoir transmissibility. Fracture closure time and transmissibility are inversely related; a higher closure time indicates lower transmissibility, and conversely, a lower closure time indicates higher transmissibility. This relationship suggested that a fracture with a longer closure time is likely to have lower permeability, resulting in slower fluid leak-off into formation, and/or higher formation fluid viscosity, which increased the resistance to displacement of pumped fluids during DFIT. In most cases, the observed closure time is influenced by a combination of both factors, where permeability and viscosity co-contribute at different magnitudes to a variable closure time. After stimulation, all wells exhibited post-stimulation flowback testing to acquire key well performance measurements, which were then used to calculate PI. Afterwards, PI was normalized per foot of lateral for all wells to account for any changes with variable lateral lengths. Once normalized PI was calculated, a cartesian plot of normalized PI versus closure time for all wells within the same spatial area was plotted to identify potential correlations. It was observed that an exponential relationship existed between the two parameters within the same localized area (Application Example 1), with an accuracy that exceeded 99%. Once the correlation was identified, the same test was implemented in another data set from a different area to confirm its repeatability (Application Example 2), which yielded similar findings. After that, additional wells with variable subsurface properties were included into the study to assess the extent of the correlation, and investigate the impact of outliers. These cases are later highlighted in Application Examples 3 and 4. The outcomes of this approach validated the feasibility of using closure time, a proxy to transmissibility, to qualitatively predict future well performance.

Application Examples

Four different application examples were investigated within the scope of this paper. The first example included the identification of a correlation between closure time and PI using four wells within the same area. The second example confirmed the repeatability of the correlation using four other wells from a different area. The third example investigated the implications of having a lower reservoir pressure on the correlation identified from the first example. Lastly, the fourth example investigated the impact of acquiring DFIT results from a section of the lateral that targeted a different localized layer which did not represent the transmissibility of the whole well.

Application Example 1 (Correlation Identification)

Four distinct wells located within the same localized area, all of which drilled and completed using identical practices and targeting the same source rock formation were analyzed. This setup allowed for the isolation of formation properties as the most dominant variable influencing well performance, enabling the use of transmissibility – or a proxy for transmissibility – to accurately predict the relative performance ranking of the four studied wells. Given the test nature of having identical DFITs across all four wells, where the volume of fluid pumped and the monitoring duration held constant, it was expected that any differences in closure time would be primarily influenced by reservoir and rock-related properties. Pre-stimulation DFIT results indicated that all four wells exhibited comparable closure pressure gradients, which indicated that all had similar geo-mechanical properties. Similarly, post-closure linear reservoir pressure gradient was identical, highlighting that the reservoir's capability to produce hydrocarbons was similar. These results were expected given that all four wells were landed in the same geological formation. However, closure time for all four wells had a remarkably wide range, with a magnitude of difference that reached 7 folds. When ranked in ascending order, the results revealed that Well-1A had the shortest closure time, while Well-1D had the longest closure time as summarized in Table-1. The pronounced variation in closure times among wells 1A, 1B, 1C, and 1D implies that factors such as fluid viscosity or formation permeability (or a mixture of the two parameters) may be contributing to the observed variability. This was further investigated by accounting for the condensate-to-gas ratio (CGR), which served as a proxy to formation fluid viscosity. Differences in CGR were noticed but were not at the level where it might hinder the outcomes of the assessment. These preliminary findings highlighted that closure time within the same area varied significantly, which indicates that transmissibility will in turn vary, and hence, variations in post-stimulation well testing were expected. Initially, wells with shorter closure time were expected to have higher transmissibility, and ultimately better well performance. Post-stimulation deliverability testing for these wells revealed a strong correlation between PI and closure time, as evident from Figure 1, which demonstrated a cartesian plot of normalized PI hydrocarbon rate, per unit of pressure, per foot of lateral versus closure time. An inverse correlation between normalized PI and closure time was clearly observed with over 99% of accuracy. Ranking these wells from highest to lowest PI inversely overlapped the closure time ranking. Well-1A, with the shortest closure time, corresponded to the highest PI, and conversely, Well-1D, with the longest closure time, resulted in the lowest PI. Consequently, this correlation analytically supported the theoretical understanding that closure time from DFIT can be used as a reliable analog to qualitatively predict well performance, and rank wells within the same area of investigation. Building on the findings of this example, the workflow was then applied on another area, with the objective of confirming its wider applicability.

Table 1—Application example 1 summary table including closure time, performance prediction, and PI ranking

Well	Closure Time Ranking	Performance Prediction Ranking	PI Ranking
Well-1A	1st (Lowest Closure Time)	1st (Best)	1st (Best)
Well-1B	2nd	2nd	2nd
Well-1C	3rd	3rd	3rd
Well-1D	4th (Highest Closure Time)	4th (Least)	4th (Least)

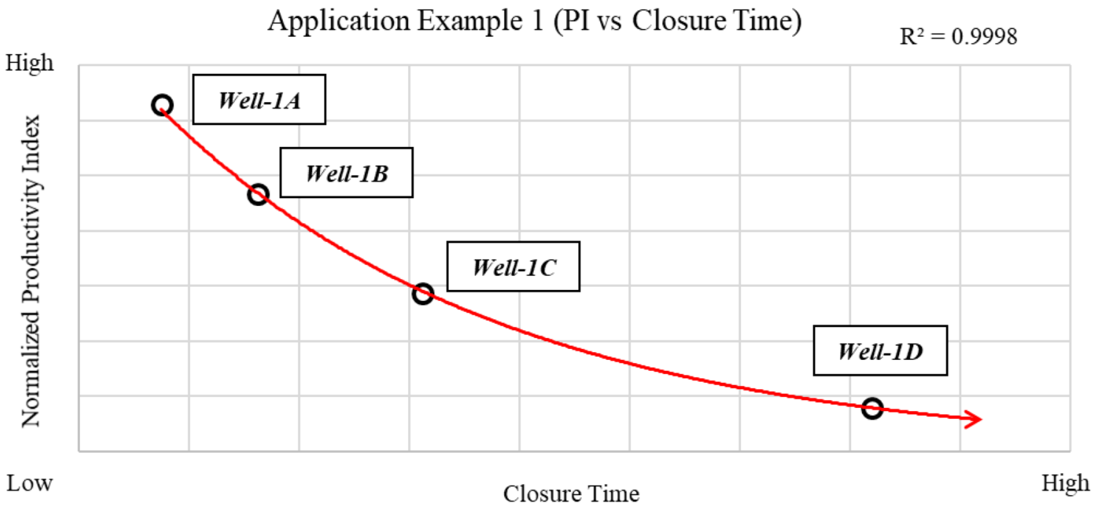

Figure 2—Application example 1 correlation between PI and closure time

Application Example 2 (Correlation Validation)

Following the successful demonstration of the first application example, the second example intended to further validate the findings and widen the applicability of the correlation using a data set from a different area. Similarly, four different wells from various locations within the same alternate area were selected as subjects of this study. All four wells underwent the same operational cycle as outlined in Application Example 1. These wells had identical drilling and stimulation practices, and were targeting the same source rock formation. Similar to Application Example 1, DFIT results from this subsequent data set yielded similar closure pressure gradients and pore pressures, yet revealed distinct closure times with over 4 folds of difference. Wells were ranked from lowest closure time to highest, where Well-2A with the lowest closure time was predicted to have the best performance, while Well-2D with the highest closure time was expected to be the lowest performer as summarized in Table-2. Plotting normalized PI versus closure time yielded a similar correlation as illustrated in Figure 3, demonstrating consistency in findings with an accuracy of over 98%. This high degree of correlation demonstrated that the relationship between PI and closure time was robust and applicable across various areas targeting the same formation, thereby supporting the credibility of using closure time as a reliable proxy for transmissibility, and consequently, a tool to predict and qualitatively compare well performance. With such consistent results, it can be concluded with a high degree of confidence that these parameters were reliable predictors of well performance in wells that targeted the same formation, and has potential to support decision-making processes and advise on potential well development strategies.

Table 2—Application example 2 summary table including closure time, performance prediction, and PI ranking

Well	Closure Time Ranking	Performance Prediction Ranking	PI Ranking
Well-2A	1st (Lowest Closure Time)	1st (Best)	1st (Best)
Well-2B	2nd	2nd	2nd
Well-2C	3rd	3rd	3rd
Well-2D	4th (Highest Closure Time)	4th (Least)	4th (Least)

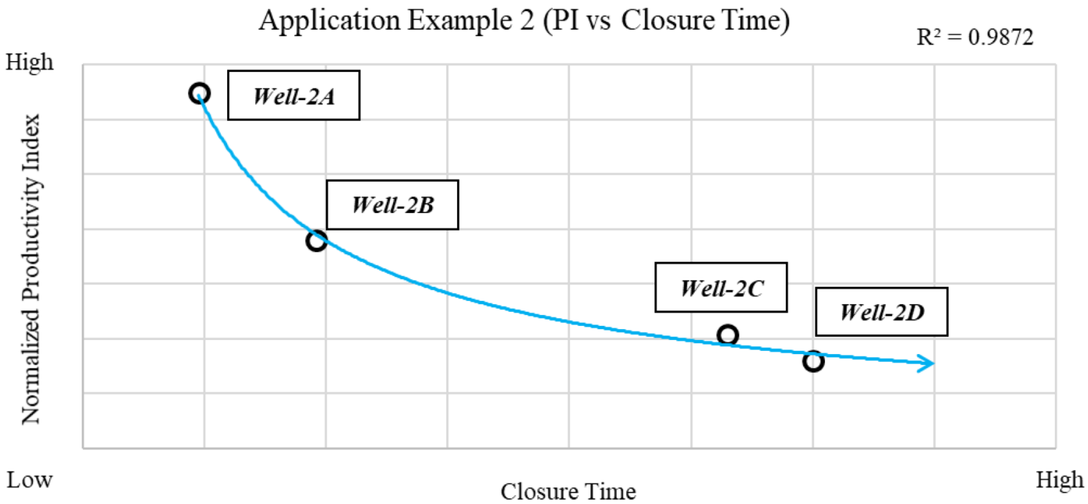

Figure 3—Application example 2 correlation between PI and closure time

Application Example 3 (Correlation Outlier – Variable Reservoir Pressure)

Utilizing the data from Application Example 1, a fifth well with lower reservoir pressure was added to the correlation. The objective of this third application was to investigate the impact variable reservoir pressure has on the correlation, and identify means to account for that variability and neutralize its impact. When the fifth well was included to the normalized PI and closure time correlation, the 99.99% correlation accuracy was reduced to 94.9%. This was attributed to the fact that the fifth well did not match the correlation trendline. To address that outlier, normalized PI for each well was multiplied by its corresponding reservoir pressure, which brought the data of the fifth well closer to the trendline, increasing the correlation coefficient to 98.5% as indicated in Figure 4.

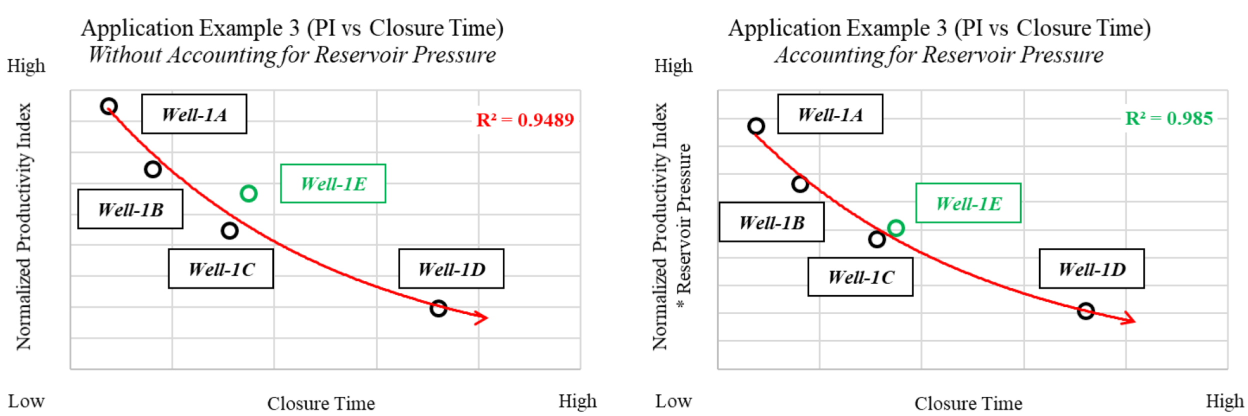

Figure 4—Application example 3 correlation between PI and closure time with and without accounting for variable reservoir pressure

Application Example 4 (Correlation Outlier – Unrepresentative Data Point)

The fourth application example builds on the learnings of Application Example 2. A fifth well with added to the data set used in the analysis, and was located in the same area. DFIT performed in the subject fifth well was in the deeper section of the lateral that had different subsurface qualities compared to the rest of the lateral, and hence, was not expected to represent the overall transmissibility of the well. Adding that fifth data point to the correlation significantly impacted its accuracy, reducing it to less than 50% as illustrated in Figure 5. This outcome highlighted the correlation's high sensitivity to outliers and unrepresentative data, even when the data point was from the same geographical area. This highlighted the importance of building the correlation on consistent and accurate to avoid misleading conclusions.

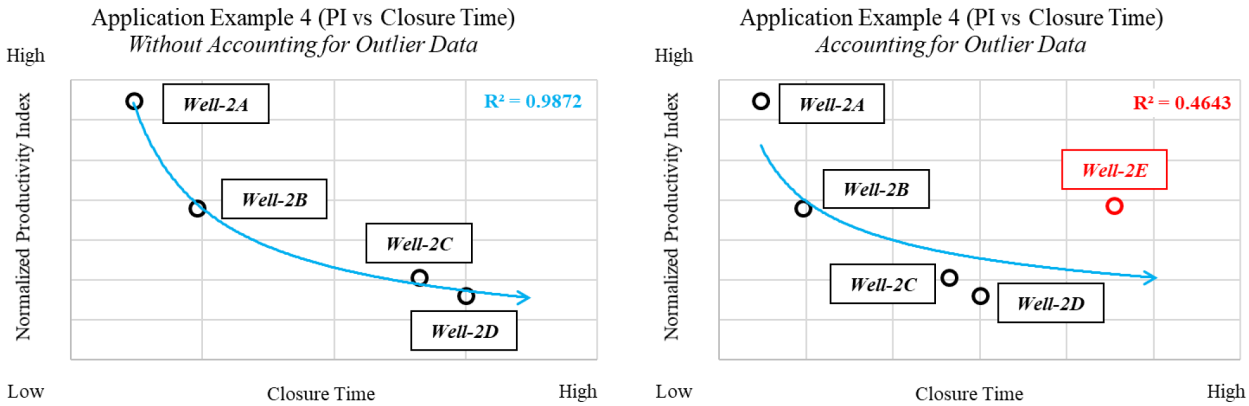

Figure 5—Application example 4 correlation between PI and closure time with and without accounting for outlier data

Results and Discussion

DFIT-driven parameters, primarily closure time, was utilized to qualitatively assess the performance of unconventional wells post-stimulation. The selection of closure time as a correlating parameter stemmed from its uniqueness to serve as a proxy to formation transmissibility, a critical parameter that directly influences well performance. Throughout the different examples discussed, a cartesian plot of PI versus closure time was used to identify the existence of a correlation between the two parameters. The evaluation was conducted on four distinct application examples, which provided the following series of understandings: (1) correlation identification, between closure time and PI; (2) correlation verification using an independent data set from a different area; (3) an assessment of the impact of variable reservoir pressure on the correlation; and (4) an evaluation of the impact of unrepresentative data on the accuracy of the correlation.

Application Example 1 summarized the outcomes of a applying the methodology on four wells located in the same geographical area, having similar subsurface properties. Closure time from all wells was obtained using DFIT, and was then correlated with PI retrieved from post-stimulation well testing. A cartesian plot of PI versus closure time was fitted with an exponential trendline that yielded a 99.99% accuracy. The findings from this exampled confirmed the existence of a correlation between DFIT-driven parameters and well performance, which can be used to qualitatively assess existing and future wells within the area of study.

Application Example 2 expanded the scope of this workflow and applied it on another set of four wells located in a different area. The objective was to confirm the repeatability of findings obtained from the first example. A similar correlation was similarly identified with an accuracy that exceeded 98%, confirming the correlation's applicability across a wider areal extent.

Application Example 3 investigated the impact of including an additional well with lower reservoir pressure to the data set used in the first example. This resulted in reducing the correlation's accuracy by 5%, which was addressed by multiplying the normalized PI for each well with its corresponding reservoir pressure, bringing the fifth well's data point closer to the trendline, and subsequently increasing the correlation coefficient to 98.5%.

Application Example 4 discussed the implications of including an additional well to the data set used in the second example. DFIT data from that fifth well did not represent the whole lateral, and significantly affected the accuracy of the correlation, reducing it to less than 50%. This example demonstrated the correlation's sensitivity to outliers, where even a single data point can significantly impact the relationship between closure time and PI.

Conclusions

Unconventional resources have been a subject of continuous optimization, with pragmatism emerging as a strategic enabler. Diagnostics in the fields of subsurface understanding and performance prediction have been growing rapidly, with a shift towards more practical alternatives. This paper introduced a non-invasive novel approach to performance prediction of unconventional wells targeting the same source rock formation. It entailed the use of DFIT-driven parameters, notably closure time, to serve as a proxy for formation transmissibility. Closure time was correlated with PI estimated from post-stimulation well testing using a cartesian plot, fitted with a trendline, and yielded consistent results. Four different application examples were studied within the body of the paper. The first application example highlighted the implementation of the methodology on four wells located within the same area. Closure time and PI from this example correlated with an accuracy of more than 99%. The second application example repeated the test performed on another set of four wells located in a different area. Results from this example yielded similar findings with a correlation coefficient that exceeded 98%. The third application example highlighted the inclusion of a fifth well with lower reservoir pressure to the data set used in the first example. Outcomes from this example suggested that the effect of reservoir pressure on the correlation can be accounted for by multiplying it with PI, then plotting the resulting values against closure time. The fourth and final application example tested the sensitivity of the correlation by incorporating an additional data point that was not representative of the overall trend. That particular data point deteriorated the accuracy of the correlation, reducing it to about 46%, emphasizing on the importance of using accurate data. Collectively, this study has demonstrated the effectiveness of using closure time as a predictive parameter for unconventional well performance targeting source rocks, through a correlation between closure time and PI. It also showcased the importance of data quality and relatability in maintaining the accuracy and reliability of the correlation.

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